Data Management Project

ENVIRONMENTAL AND SEASONAL EFFECTS ON FOOTBALL MATCH PERFORMANCE: ANALYSIS OF THE BIG FIVE EUROPEAN LEAGUES OVER THE LAST THREE SEASONS

Master's Degree in Data Science – Academic Year 2025–2026
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INTRODUCTION

Modern football relies on data-driven approaches for performance evaluations. However, the influence of **environmental factors,** such as weather and seasonality, remains relatively underexplored. Existing research primarily focus on technical or tactical indicators, often relegating meteorological variables to a secondary role.

OBJECTIVE

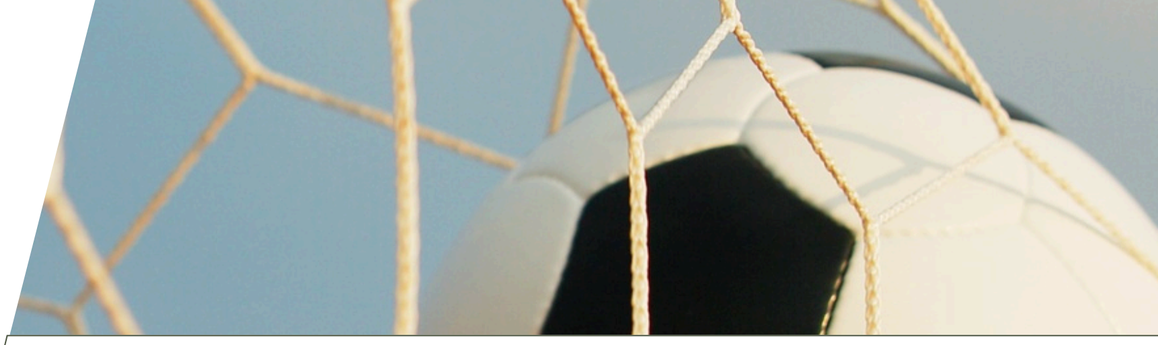
The **goal** is to investigate the relationship between weather conditions, seasonal effects, and soccer match performance.

RESEARCH QUESTIONS

- Does **weather** influence match outcomes and match statistics?
- Do **seasonal effects** influence match outcomes and match statistics?

MODULAR PIPELINE DESIGN

The project follows a **modular** and **pipeline-based architecture** design to ensure maintainability and reproducibility.



DATA ACQUISITION & ENRICHMENT

Collects data through **web scraping** of FBref and integrates additional information through **external API** (OpenCage Geocoding API e Open-Meteo Historical Weather API).

DATA STORAGE

Provides a structured representation of the data leveraging **PostgreSQL** (18.1) with **PostGIS** for spatial and relational data.

ANALYSIS & VISUALIZATION

Includes **exploratory data analysis**, **statistical testing** and **visualizations**.

DATA QUALITY

Focuses on assessing and improving the reliability of the dataset through **validation checks** and **corrective strategies** to identify and handle data quality issues.

SOFTWARE ARCHITECTURE



SCRAPER FOR FBREF

(https://fbref.com/en/comps/)

STRUCTURE

The scraper follows an **automatic hierarchical navigation strategy**: it starts from the competition-level page, extracts the Top-5 leagues table, and iterates over the selected seasons. For each season, it accesses the Scores and Fixtures page and collects links to individual match reports.

EXTRACTED DATA

Each match report page provides detailed team statistics, including **match info** (teams, date, time, score, xG, stadium, attendance) and **performance metrics** (possession, passes, shots, saves, cards, and many others).

5,330 matches

Big 5 European leagues over the last three completed seasons (2022/2023, 2023/2024, 2024/2025)

DATA ACQUISITION AND ENRICHMENT

DATA ACQUISITION AND ENRICHMENT

SEASONALITY

Seasonality was captured by enriching the data with **meteorological seasons** (summer, autumn, winter, spring) and **time slots** (midday, afternoon, evening, night).



WEATHER CONDITIONS

Weather data required several preliminary steps before calling the API:

- Stadium coordinates were obtained using the **OpenCage Geocoding API** (latitude, longitude, country)



- **Timezones** were identified to align match kick-off times with the corresponding weather data



OPEN-METEO HISTORICAL WEATHER API







For each match, **precipitation**, **wind speed**, and **temperature** were collected. To account for weather variability before and during the match, data were gathered within a **temporal window spanning two hours before to two hours after kick-off**.

Weather variables were then **categorized** to improve interpretability.

DATABASE STRUCTURE

6 ENTITIES

- Matches
- Weather_Conditions
- Teams
- Stadiums
- Competitions
- Match_Statistics

6 RELATIONSHIPS

- Matches – Match_Statistics
- Matches – Weather_Conditions
- Competitions – Matches
- Stadiums – Matches
- Teams (Home) – Matches
- Teams (Away) – Matches

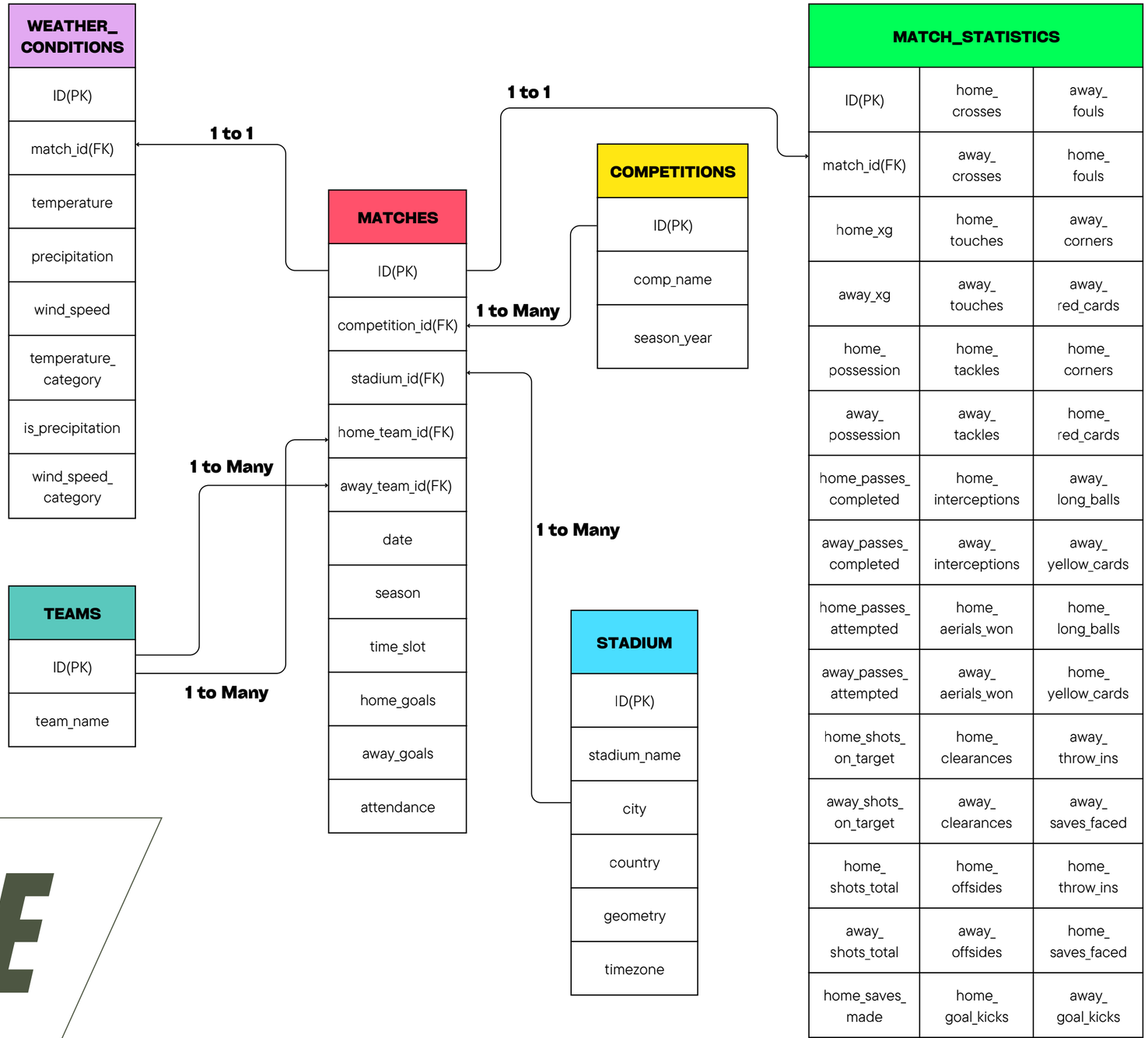
RELATIONAL DATABASE

- Entities are well-defined
- Efficient JOIN and GROUP BY for analytical queries
- Strong data integrity through constraints (NOT NULL, UNIQUE, CHECK)
- Moderate data volume

PostgreSQL (with PostGIS extension)

DATA STORAGE

Football Database



DOES WEATHER INFLUENCE MATCH OUTCOMES AND MATCH STATISTICS?

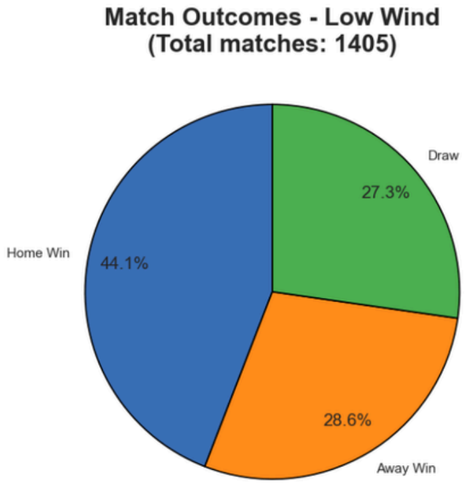
Match Outcomes

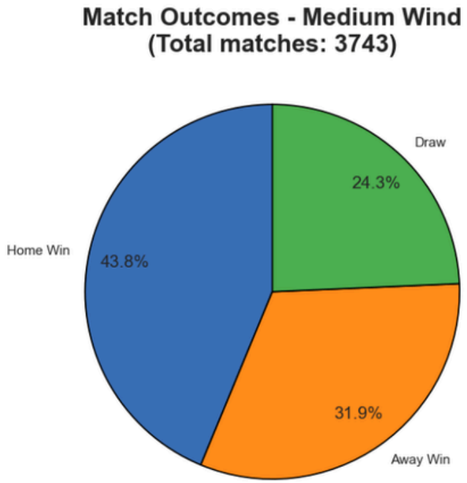
The meteorological variables considered were **precipitation**, **wind**, and **temperature**.

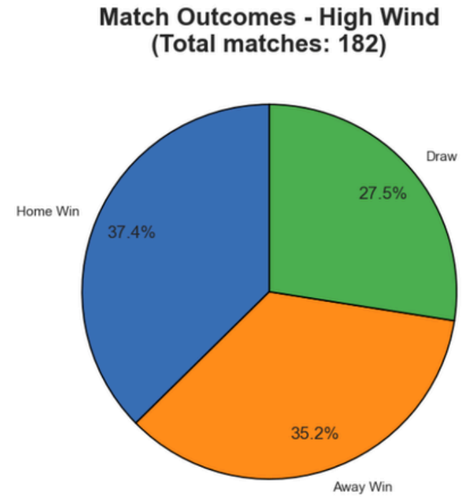
The following methodology was applied:

- Visual analysis of **pie charts**
- Construction of **contingency tables** between each meteorological variable and the match result (home win, draw, or away win)
- Application of the **Chi-square test** of independence

After applying this strategy for each factor, it was found that both **temperature** and **precipitation** do **not influence** match outcomes, as they do not show a significant effect in the dataset considered.







WIND

Unlike rain and temperature, the analysis of the pie charts immediately shows **visible differences** in the distribution of match outcomes across the different **wind categories**. The chi-square test confirms that this difference is **statistically significant** (p-value = 0.037 < 0.05), indicating that wind speed may influence the probabilities of a win, draw, or loss in the **analysed dataset.**

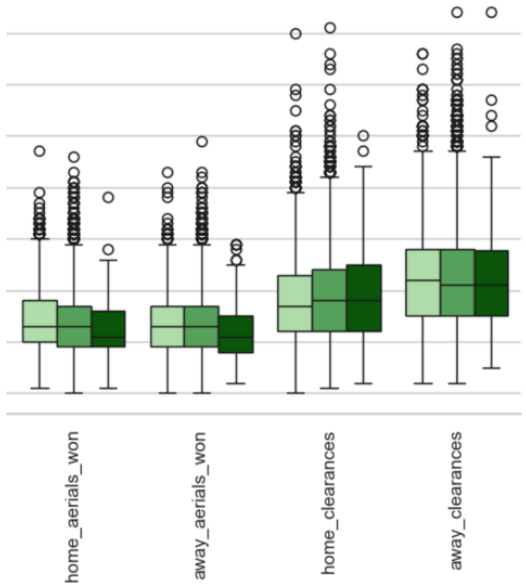
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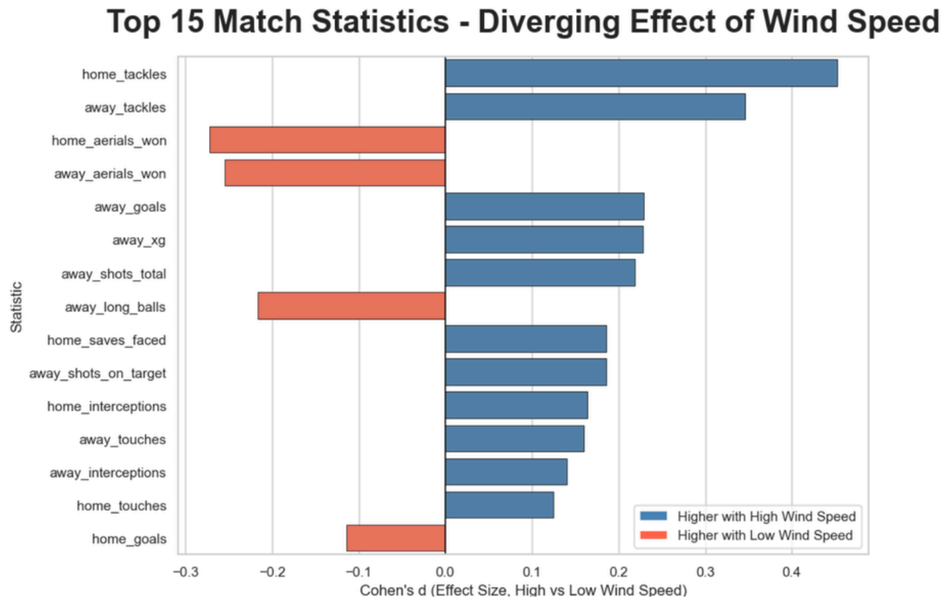
Match Statistics

To investigate whether meteorological factors influenced match statistics, the following methodology was applied:

- Visual analysis of **boxplots**
- **Cohen's** d to measure effect sizes:
  - 0.2 → small effect
  - 0.5 → medium effect
  - 0.8 → large effect
- Visual analysis of **diverging chart**

After applying this strategy for each factor, it was observed that **precipitation** has a **negligible** effect, with Cohen's d values below 0.2. In contrast, **temperature** shows values approaching 0.3, indicating a **moderate effect**. In particular, under **hotter** conditions, matches tend to be **slower** and more **interrupted** in the dataset considered.





WIND

Wind showed a **stronger effect** than the other two meteorological factors. Comparing the **boxplots** statistics and **Cohen's** d values, some values around **0.5**, was found. These findings indicate that wind has a medium influence on match statistics, contributing to a more **fragmented** and **less fluid** style of play.

DO SEASONAL EFFECTS INFLUENCE MATCH OUTCOMES AND MATCH STATISTICS?

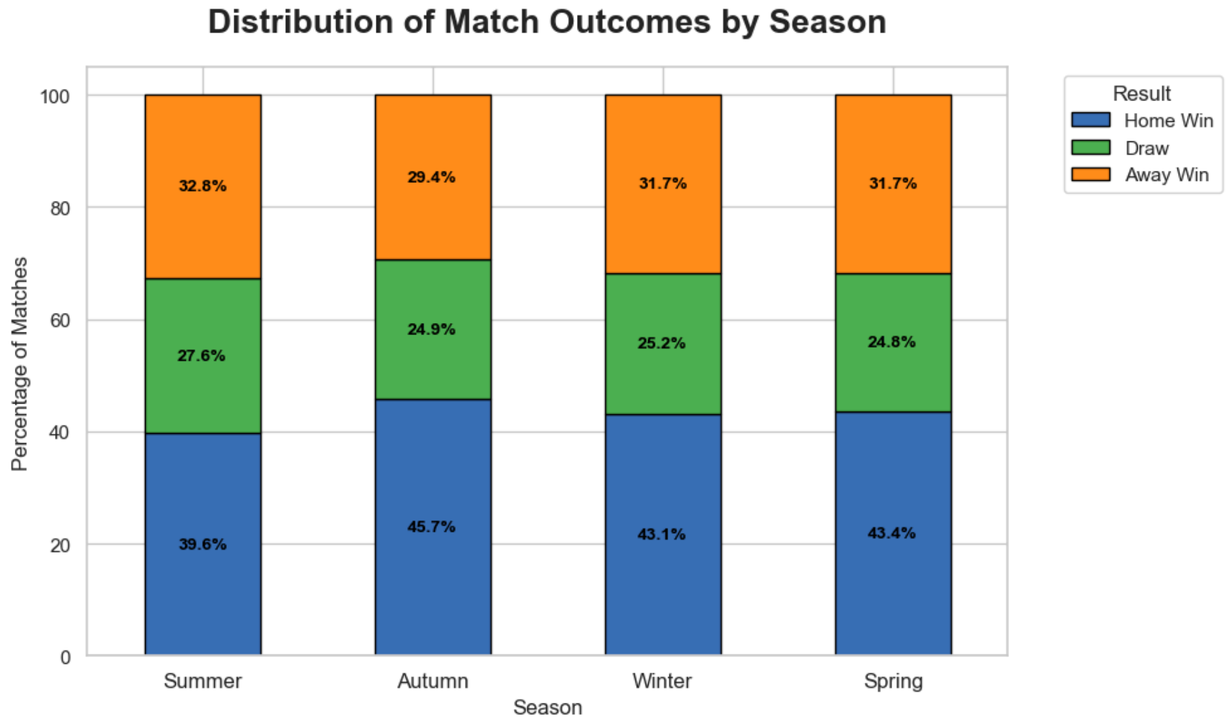
Match Outcomes

The seasons considered were **summer**, **autumn**, **winter** and **spring**. The following methodology was applied:

- **Distribution analysis** across the four seasons
- Construction of **contingency tables** between season and match result
- Application of the **Chi-square test** of independence

The chi-square test yielded a p-value of 0.325 (> 0.05), indicating **no statistically significant** evidence that season influences match outcomes.

Although **slight variations** are observed, these differences remain small in magnitude and do not demonstrate a systematic seasonal effect on home wins, draws, and away wins.



DO SEASONAL EFFECTS INFLUENCE MATCH OUTCOMES AND MATCH STATISTICS?

STATISTICAL CONCLUSION

Although seasons do not influence match outcomes, they **systematically shape match dynamics**. The transition is gradual, with the most significant differences occurring between **summer** and **winter/spring**.

Match Statistics

To investigate whether and which performance metrics differ across seasons, the following methodology was applied:

- **Kruskal–Wallis test** to compare distributions across seasons, ranked by **Epsilon-squared** to identify the most affected performance statistics
- **Dunn test with Bonferroni correction** to identify specific seasonal shifts







DEFENSIVE INTENSITY

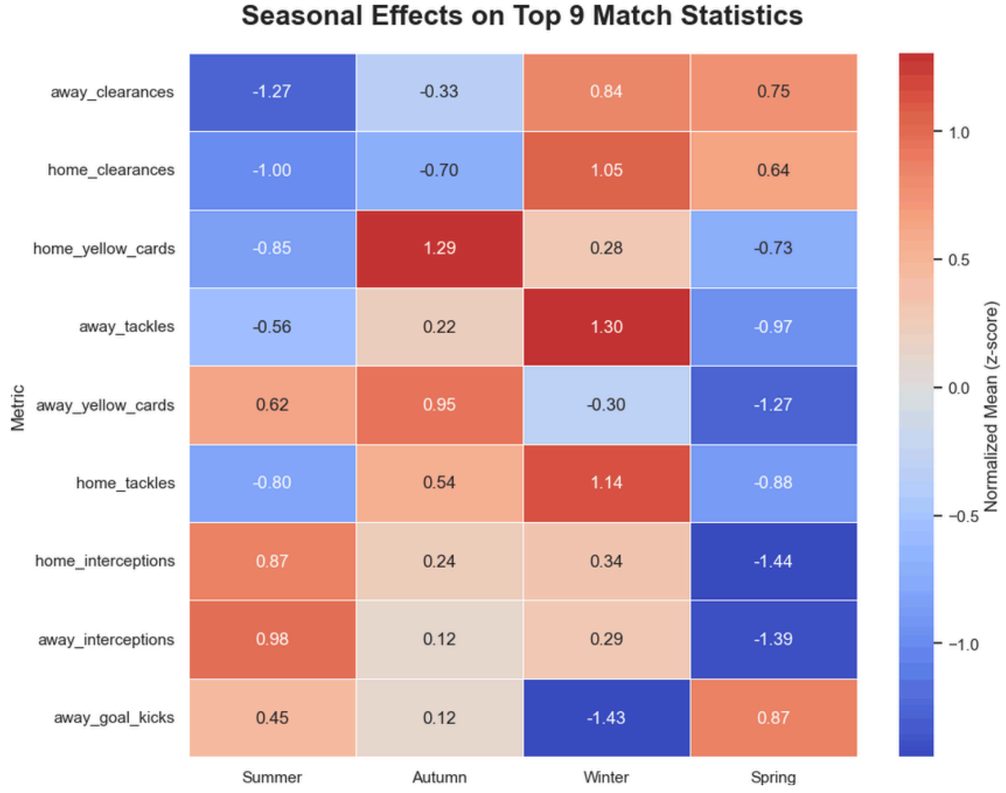
Clearances and physical engagement (tackles/interceptions) increase significantly in **winter**, suggesting a more defensive playing style.

DISCIPLINARY TRENDS

Yellow cards peak in **autumn** before declining, indicating higher early-season aggression or intensity and a more rigorous enforcement of disciplinary directives.

MATCH FLOW

Metrics like goal kicks reach their lowest in **winter**, reinforcing a shift in match dynamics during the coldest months.



ASSESSMENT

An **ex-post data quality** was intentionally performed to validate the reliability of the analysis already executed and to verify whether potential data issues could affect the interpretation of the results.

IMPROVEMENT

Given the limited number of anomalies and their strong connection to real-world events, all improvements were implemented through **database views**, ensuring data traceability and reproducibility, rather than modifying or overwriting the original data.

DATA QUALITY

Dimension	Check	Issues	Notes
Completeness	Missing attendance values	2 matches	Matches played behind closed doors
	Missing match statistics	1 match	Match suspended before completion
Spatial consistency	Stadium-country coherence	0	All stadiums correctly assigned to their corresponding country
	Overlapping stadium locations	12 pairs	Duplicate records referring to the same physical stadium due to naming variations
Logical consistency	Completed vs. attempted passes	0	Completed passes never exceed attempted passes
	Shots on target vs. total shots	0	Shots on target always less than or equal to total shots
	Disciplinary statistics	0	Yellow and red cards always non-negative
	Ball possession sum	8 matches	Possession sums slightly exceeded 1 due to rounding effects (max 1.01)
	Saves, shots on target, goals	1 match	Officially awarded match
Temporal consistency	Season-date alignment	0	All match dates within the expected season boundaries

COMPLETENESS AND CONSISTENCY

CONCLUSION

SUMMARY OF FINDINGS

MATCH OUTCOMES

Weather and seasonality have **minimal impact** on match results. Only **wind speed** demonstrates a moderate effect.

MATCH DYNAMICS

Meteorological and seasonal factors significantly shift **how** the game is played:

- **Wind**: strongest effect, affecting physical duels, tackles, and aerial play
- **Temperature**: colder conditions correlate with more physically demanding play, while heat slows match flow
- **Seasonality**: defensive metrics (clearances) increase toward winter, while intensity (yellow cards) peaks in autumn

DATABASE STRUCTURE

Successfully integrated **FBref** and **Open-Meteo Historical Weather** data into a structured **PostgreSQL** relational database with PostGIS extension for spatial analysis.



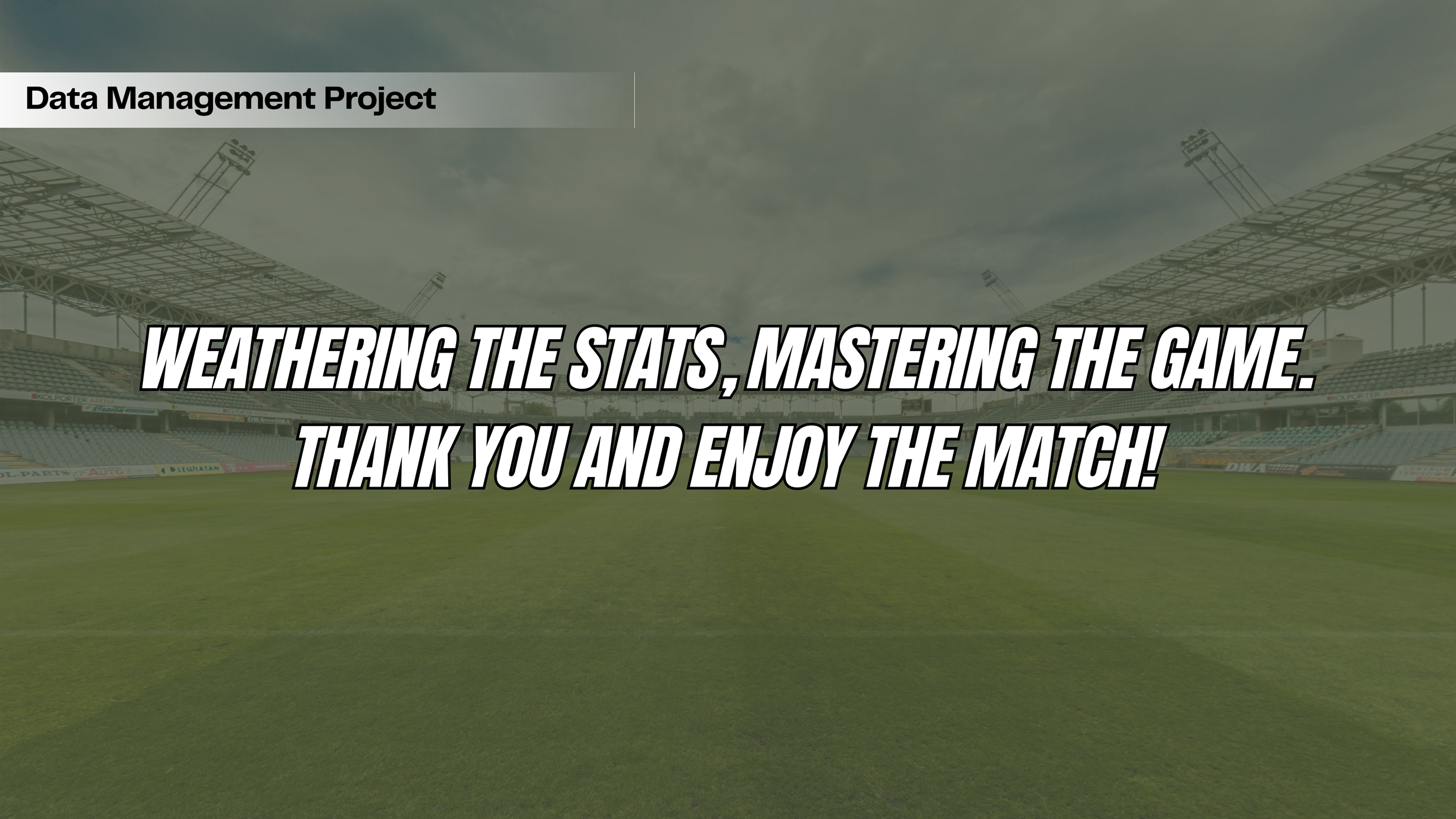
SCALABILITY

Optimize the data collection pipeline through **parallel processing** and **automated scheduling**, while scaling the dataset geographically (other continents) and temporally (more seasons) to capture **long-term and intra-day climate trends**.

ADVANCED MODELING

Move from univariate methods to **Machine Learning** (random forests/neural networks) to predict performance based on **complex environmental interactions**.

FUTURE DEVELOPMENTS



WEATHERING THE STATS, MASTERING THE GAME.
THANK YOU AND ENJOY THE MATCH!